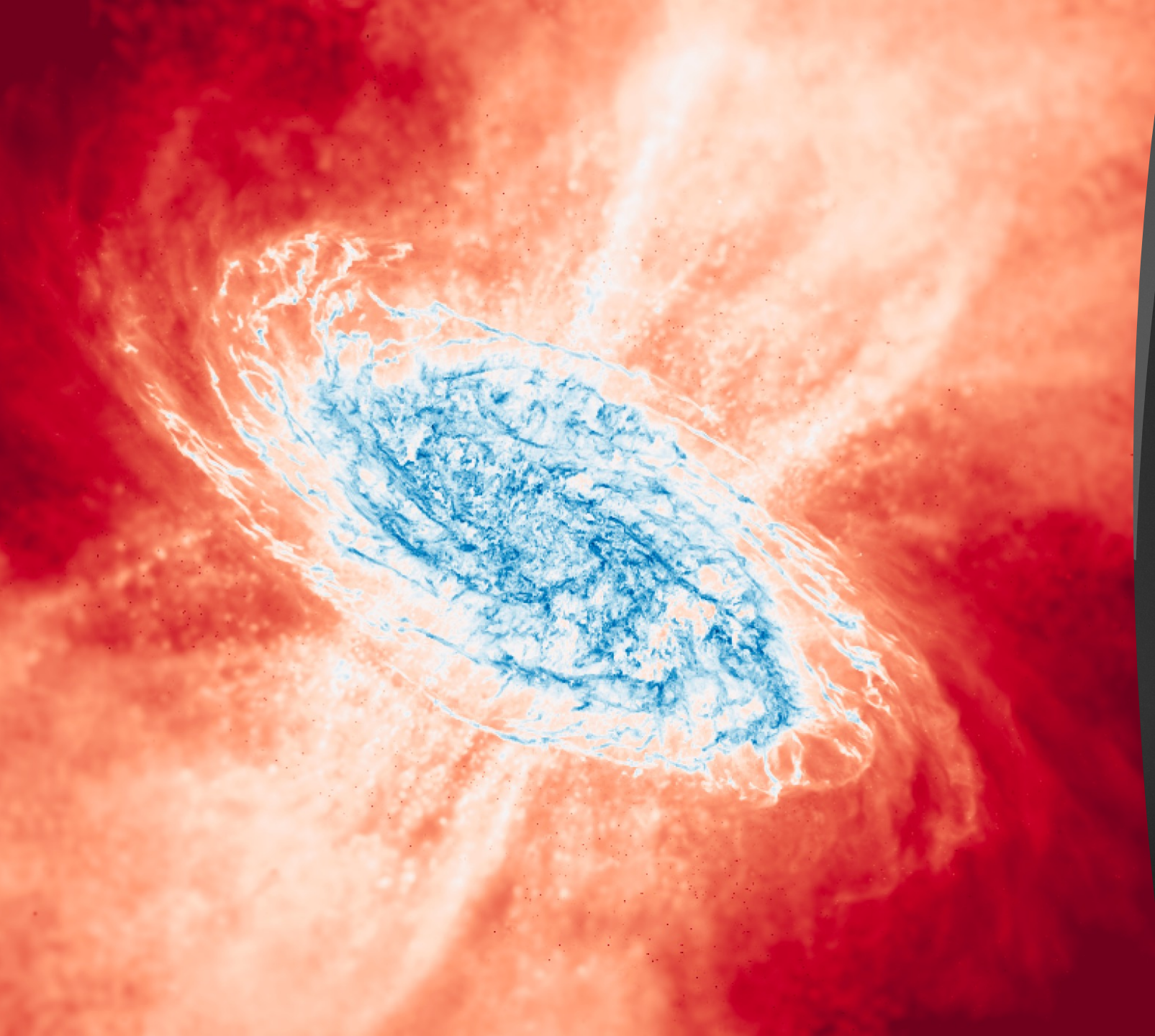
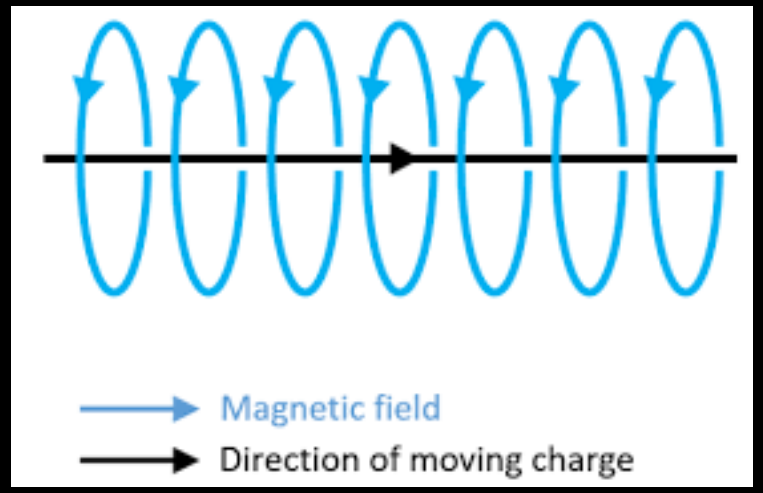


Simulating Galactic Dynamo Processes with Smoothed Particle Magnetohydrodynamics

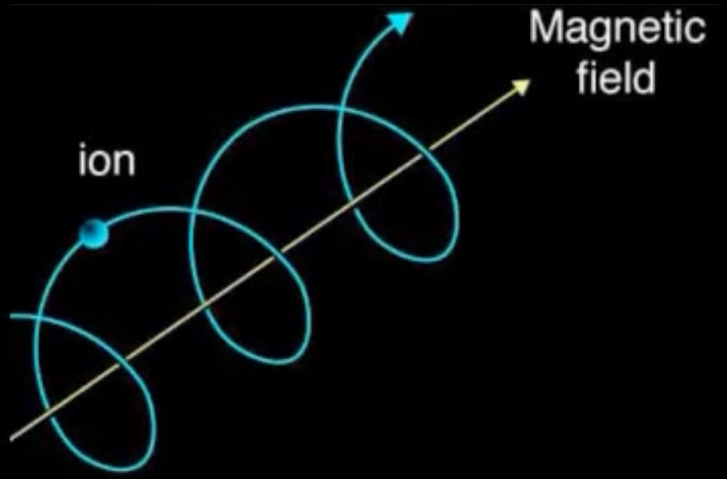
Robert Wissing

Institute of Theoretical Astrophysics, University of Oslo

Magnetic fields ⇔ charged particles



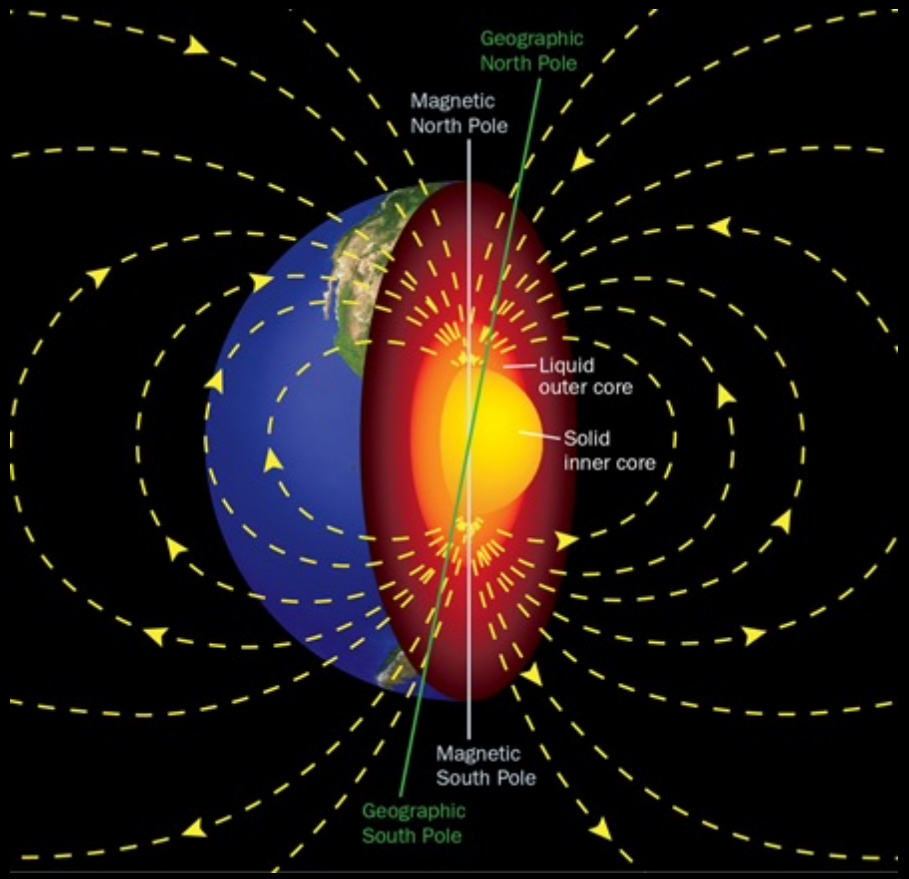
Magnetic fields are produced by the motions of charged particles

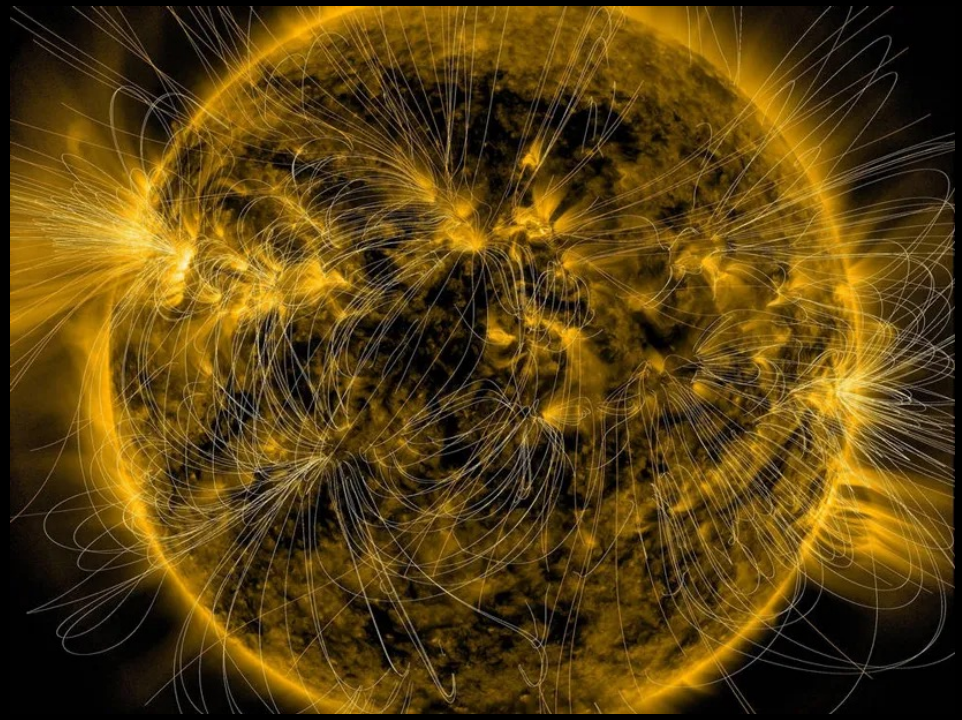


Charged particles trajectories are curved by magnetic fields

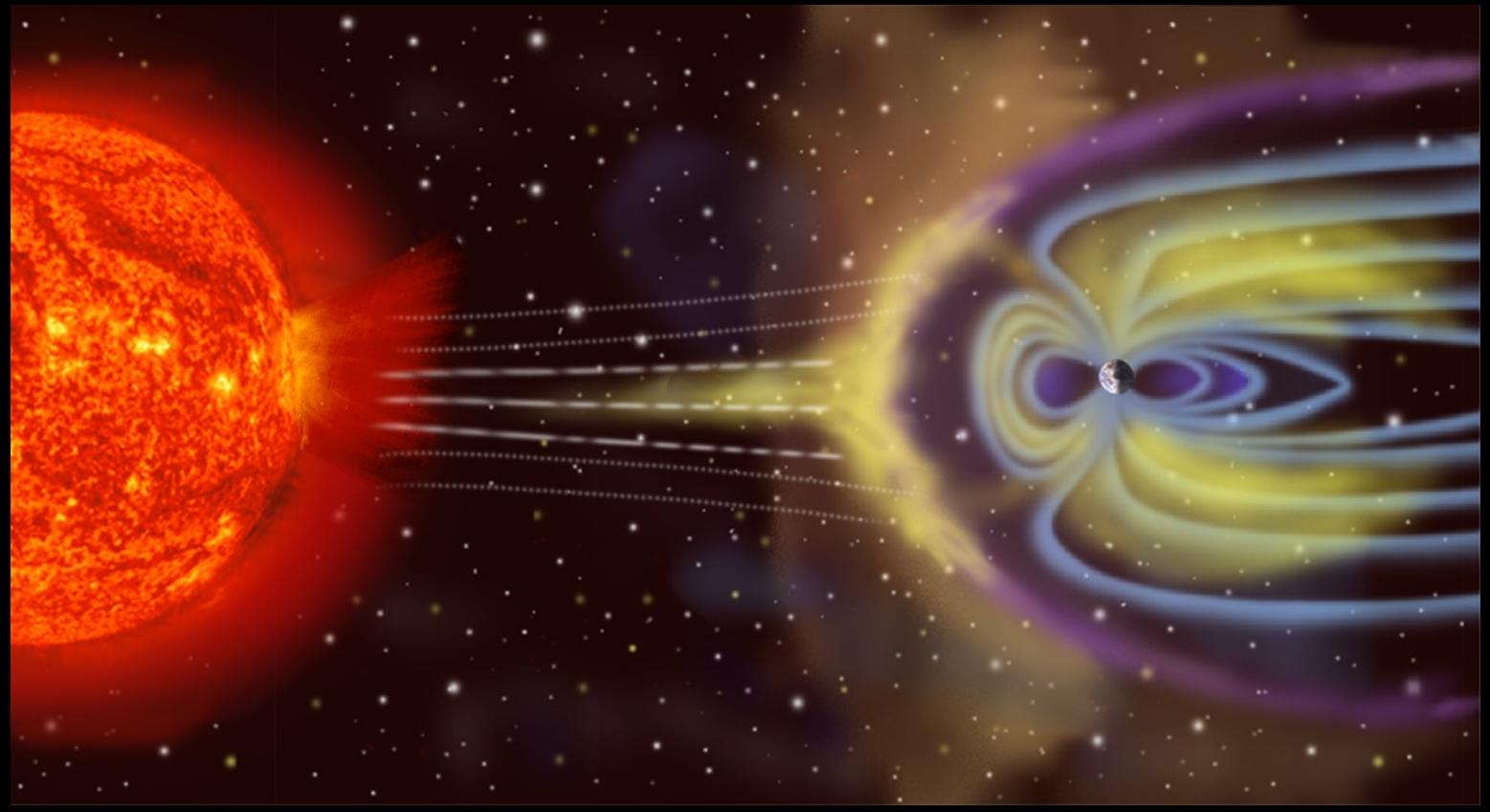
Magnetic fields and charged particles go hand in hand.

Magnetic fields in our solar system





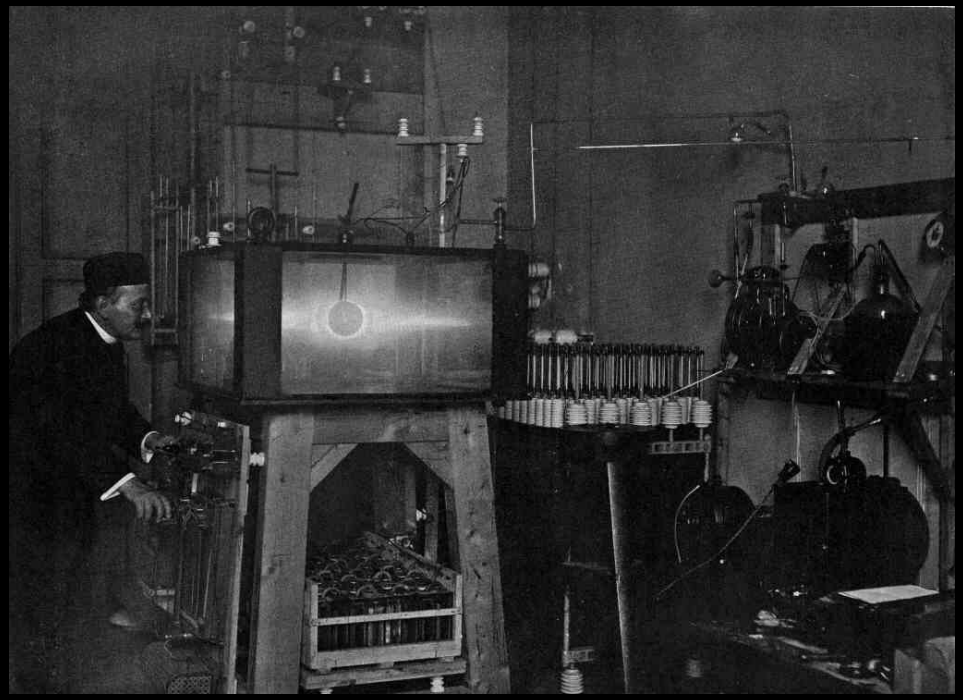
Magnetic fields in our solar system

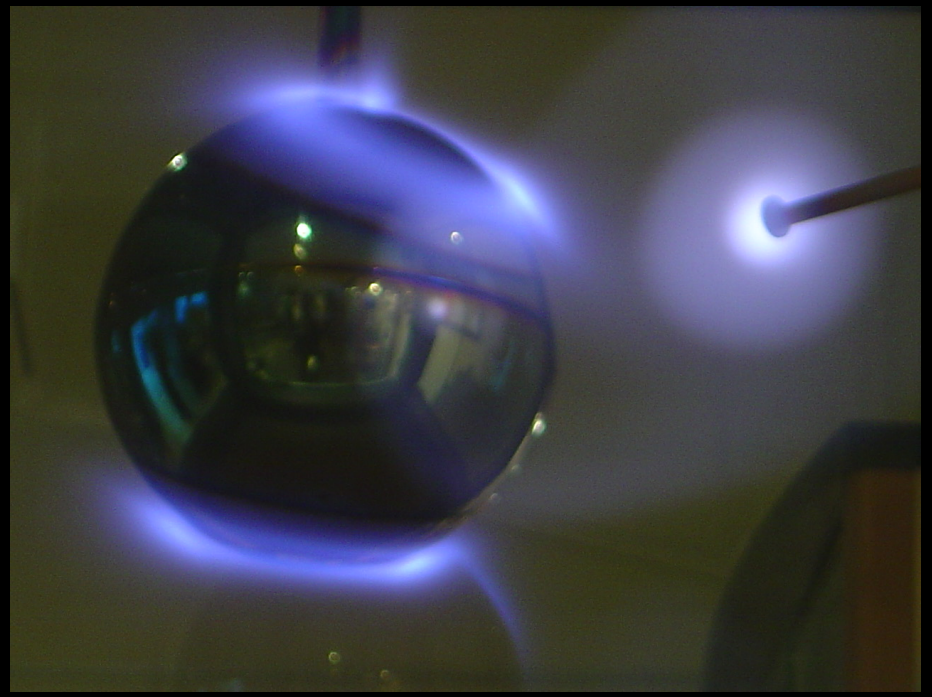


Magnetic fields in our solar system



Magnetic fields in our solar system

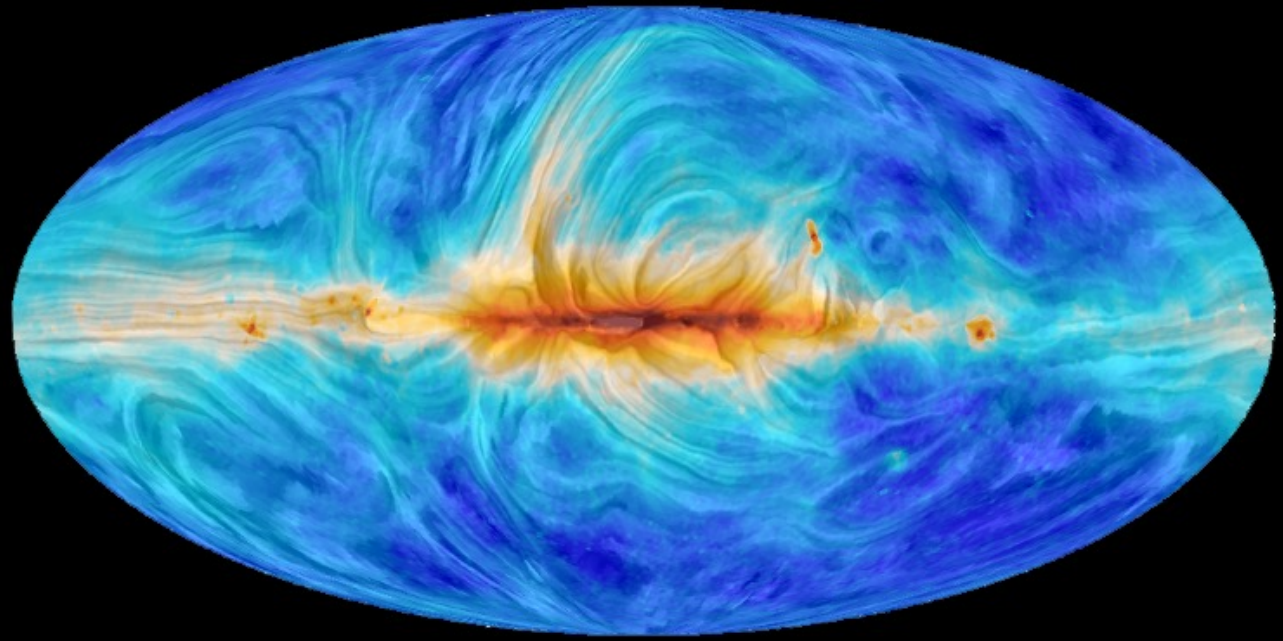




Galaxies are magnetized too!



Hubble telescope
M51 "Whirlpool Galaxy" (Borlaff et al. 2021)



Planck Satellite
Milky Way (Planck Collaboration 2016)

Most gas in galaxies contain large fraction
of charged particles

Where does galactic magnetic fields come from ?

Primordial fields
10^{-20} G



10^{14} times stronger!

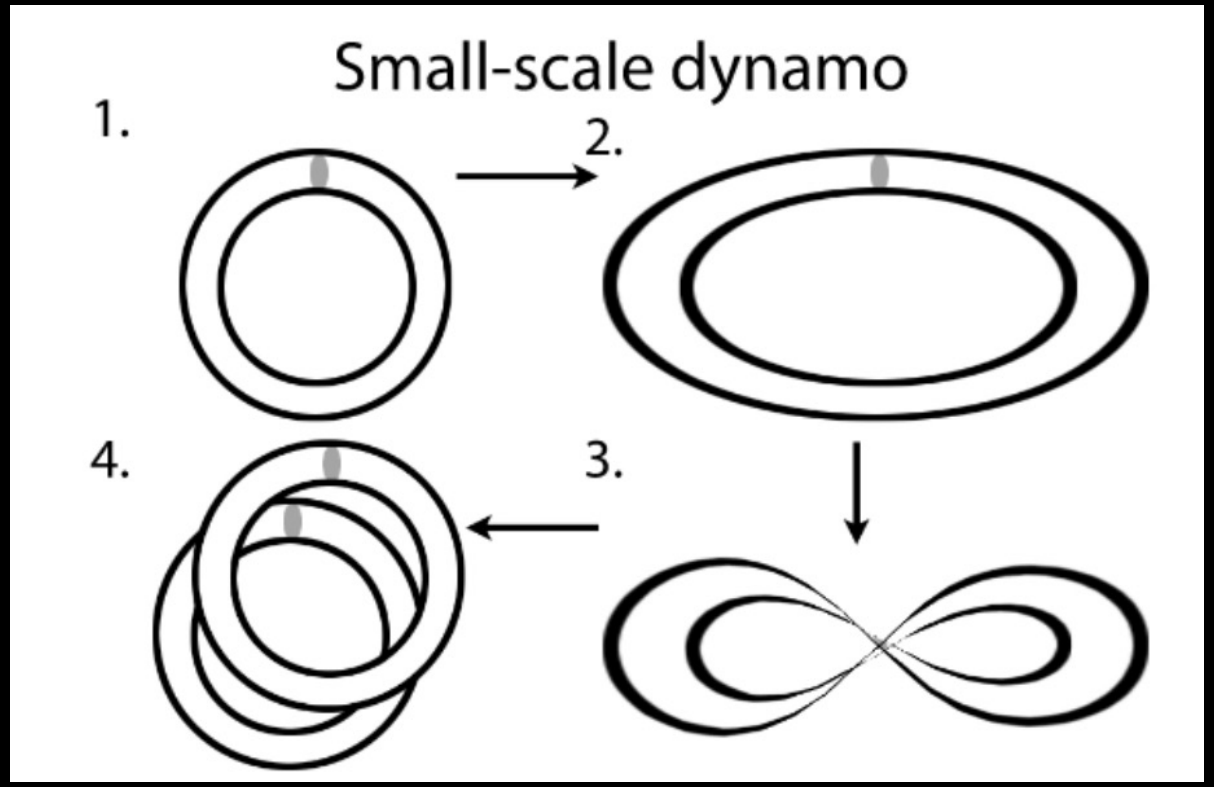
Magnetic field in Galaxies
10^{-6} G



Fast growth of magnetic field required!

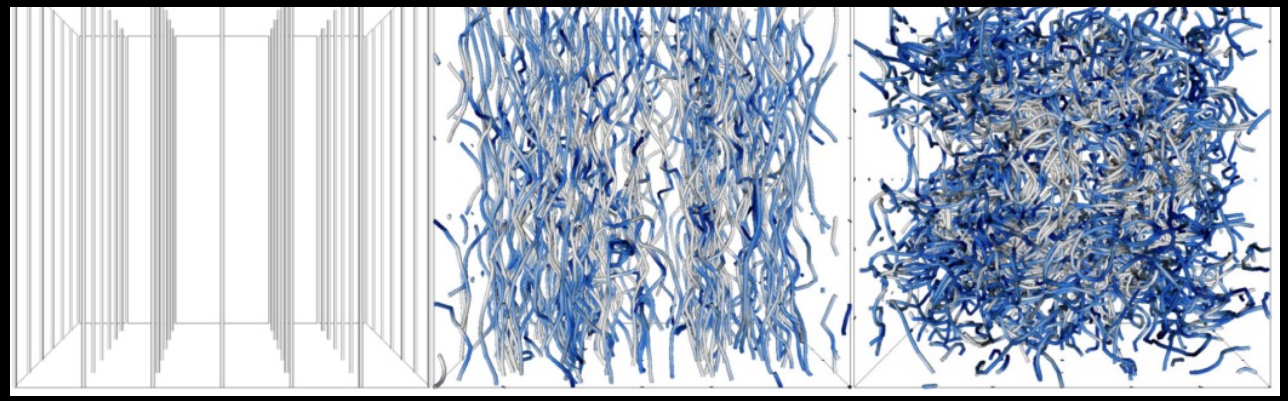
Dynamo - How magnetic fields can grow fast

- Streching, twisting and folding of field lines by fluid motions

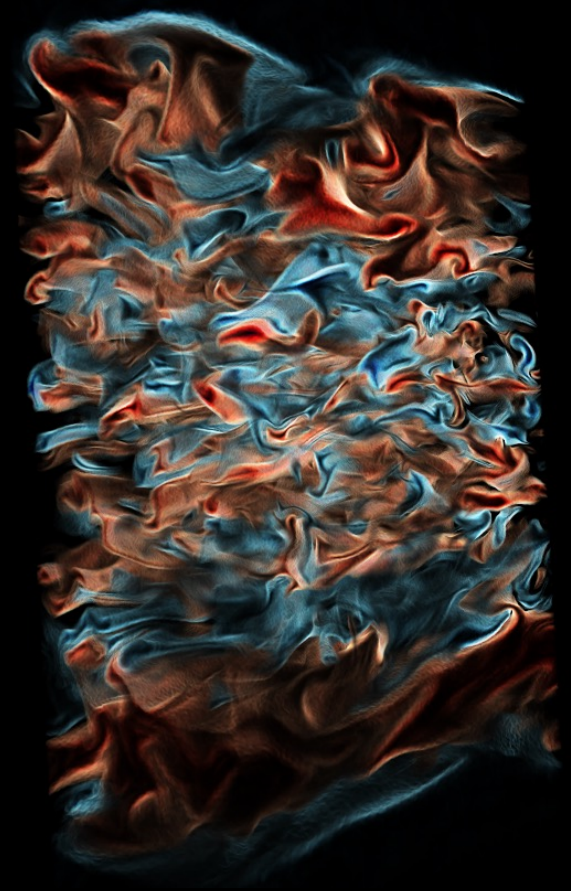


Dynamo - How magnetic fields can grow fast

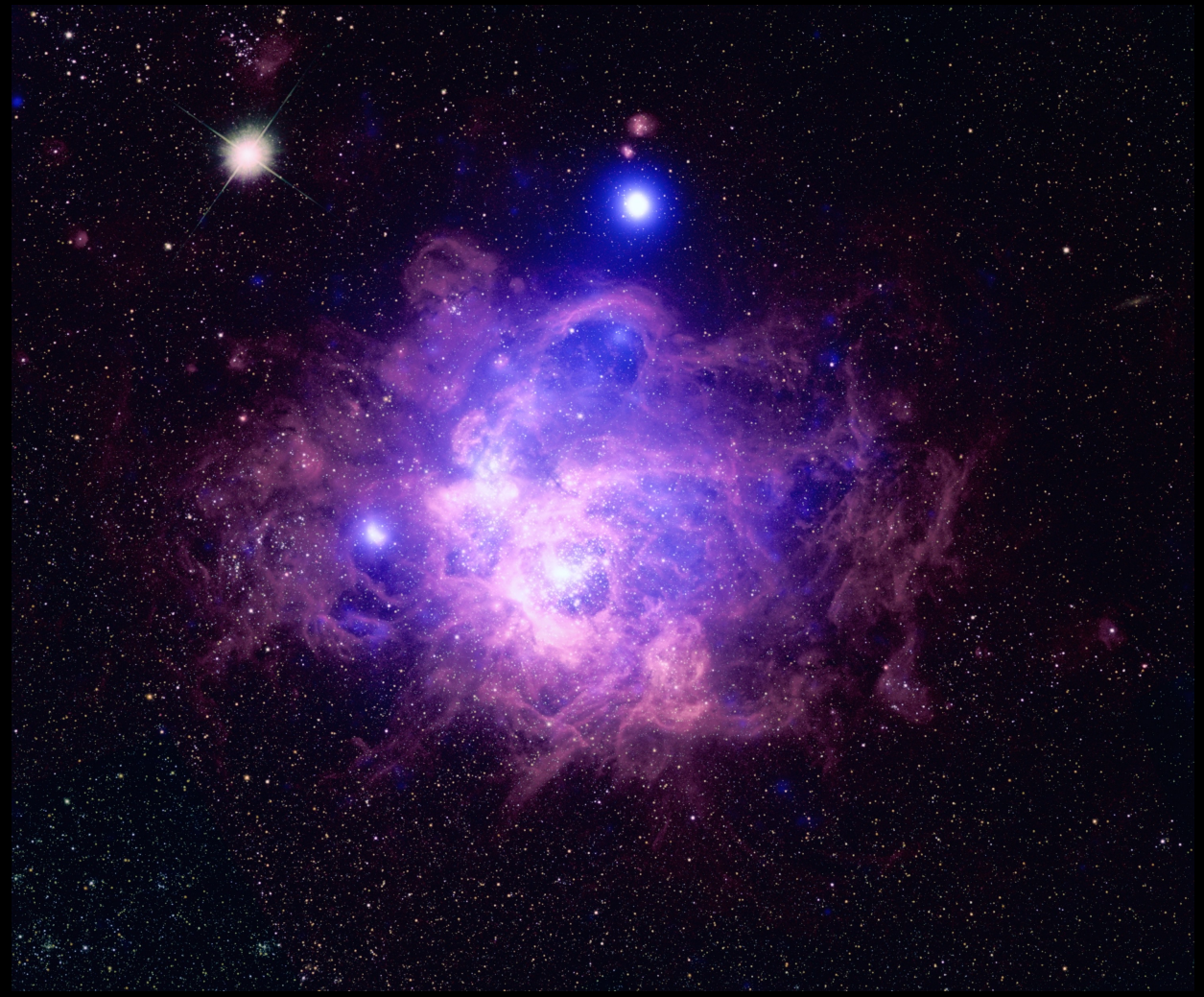
- Turbulence can stretch, twist and fold magnetic fields!
- Galaxies are highly turbulent
- Rotation in galaxies can also order magnetic fields



Federrath 2016



Role of magnetic fields in galaxies



Chandra + Hubble telescope
NGC 604

Energy content of magnetic fields can be equal or greater then the thermal energy content
-> dynamically important!

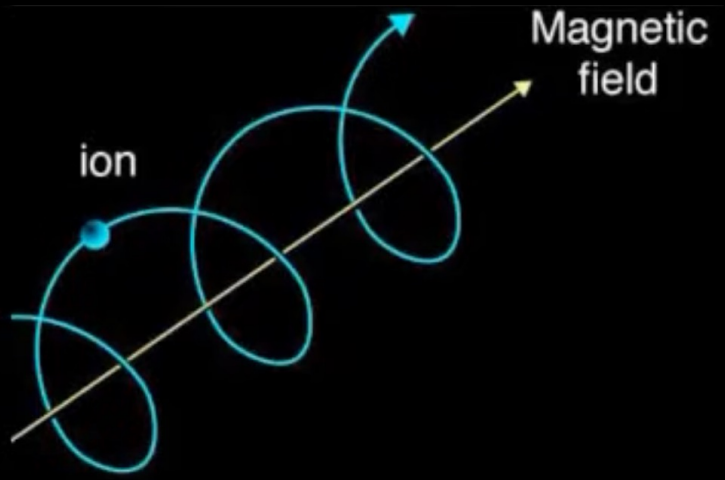
Magnetic fields can slow or stop the collapse of molecular clouds, which is the first stage in the star formation process

Role of magnetic fields in galaxies



Supernova remnant IC 443
Nasa Fermi telescope

Cosmic ray -> charged particles released from supernova moving near the speed of light throughout the galaxies



Travel along magnetic fields and potentially drives galactic outflows

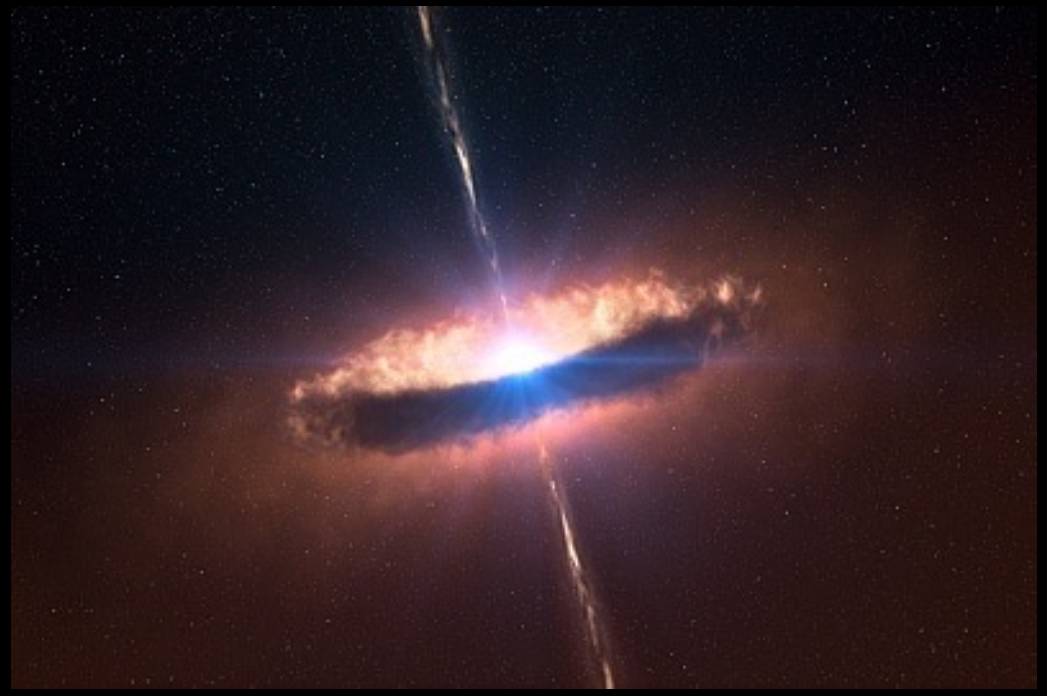
Role of magnetic fields in galaxies



EHT Collaboration
Black hole in centre of M87

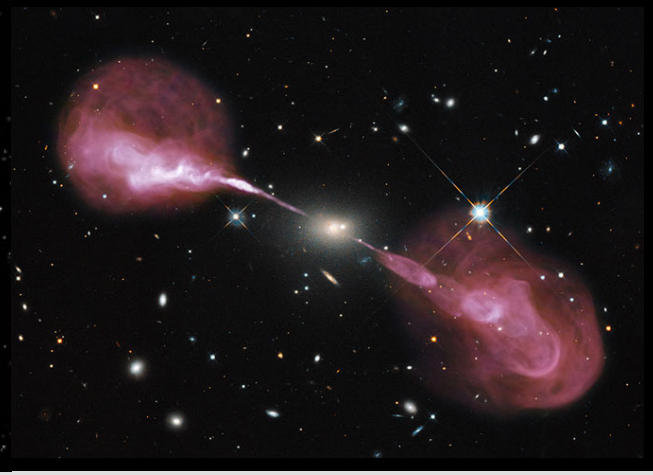
Found around accretion disks of black holes, helps in "feeding" the black holes

Role of magnetic fields in galaxies



Believed to be responsible for the formation of astrophysical jets.





Goals of the thesis work

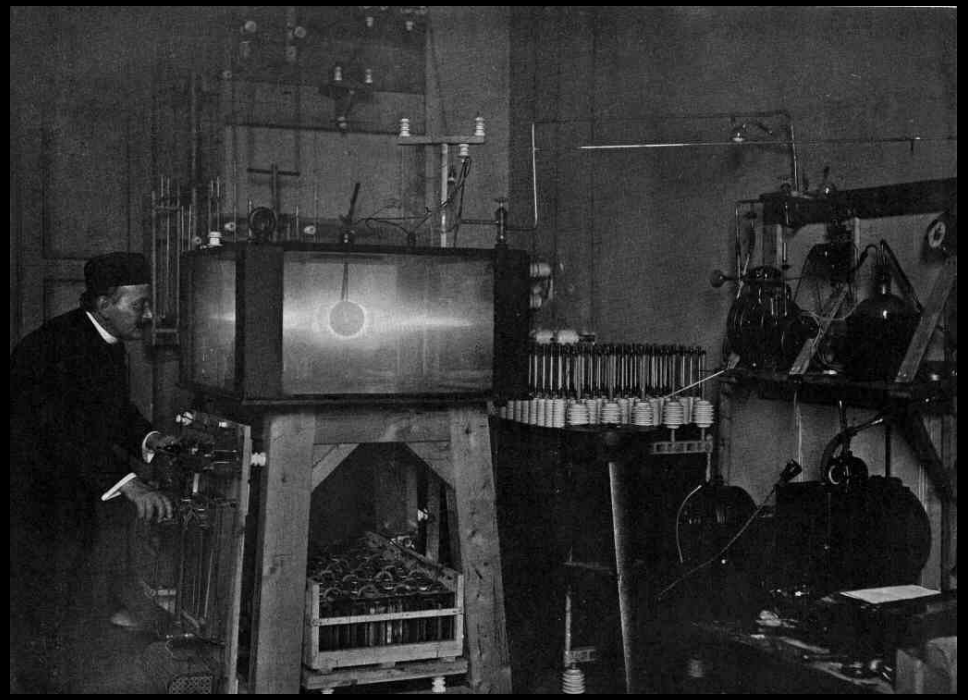
- Develop a numerical scheme for galaxy simulations that can properly model magnetic fields
- Application of the method to challenging physical processes:

Jet formation in magnetized core collapse

Capture the periodic magnetic field reversals present in accretion disks

- Explore the amplification processes present in galaxies

Creating a virtual laboratory – Numerical simulations

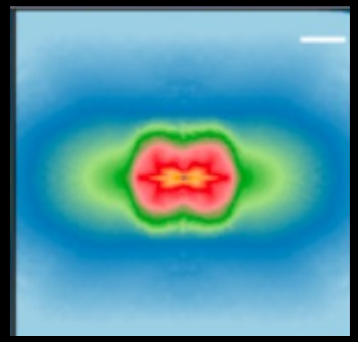




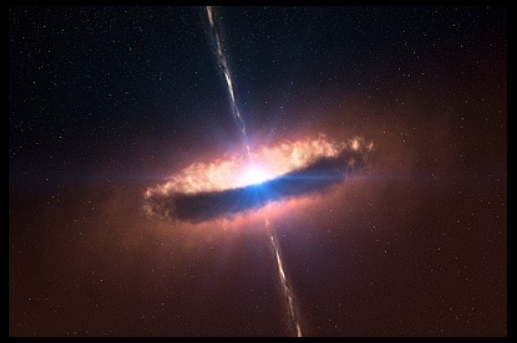
Magnetic fields are hard to model numerically

- Magnetized cloud collapse

Fails to capture jet formation!

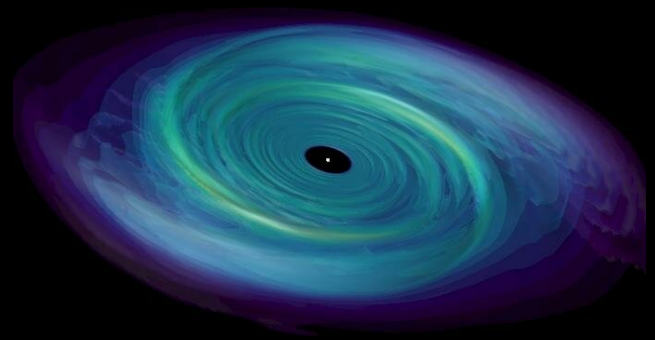


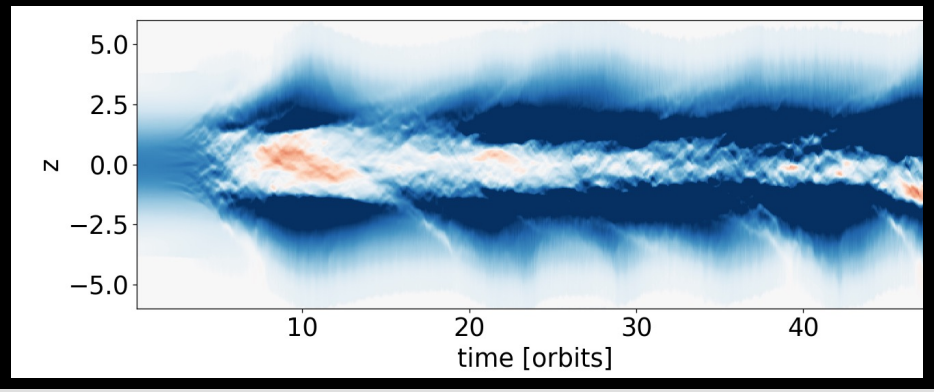
≠



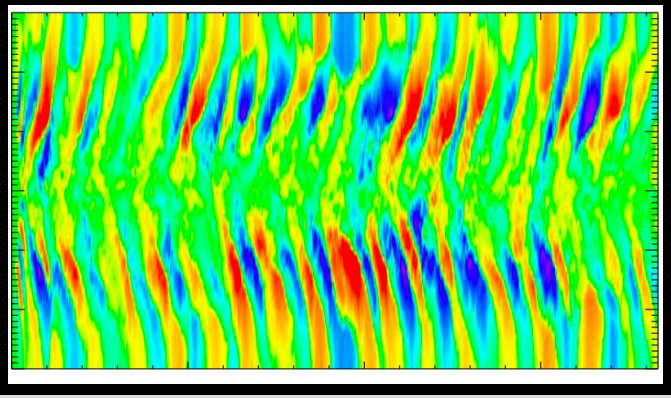
- Magnetic fields in accretion disk

Run-away growth!





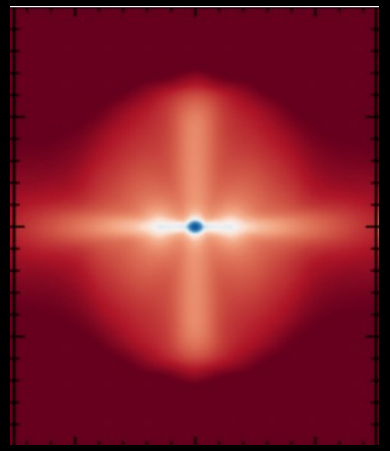
≠



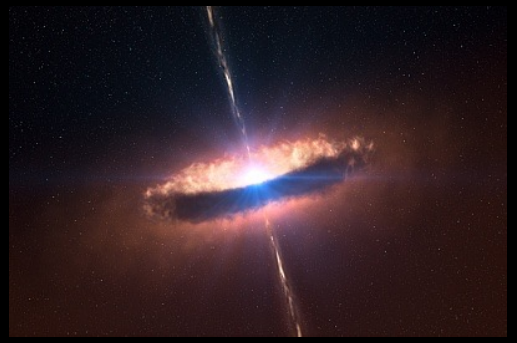
Enter GDSPH!

- Magnetized cloud collapse

Successfully captures jet formation

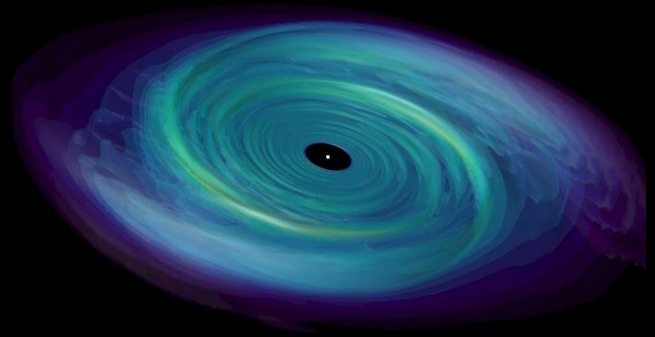


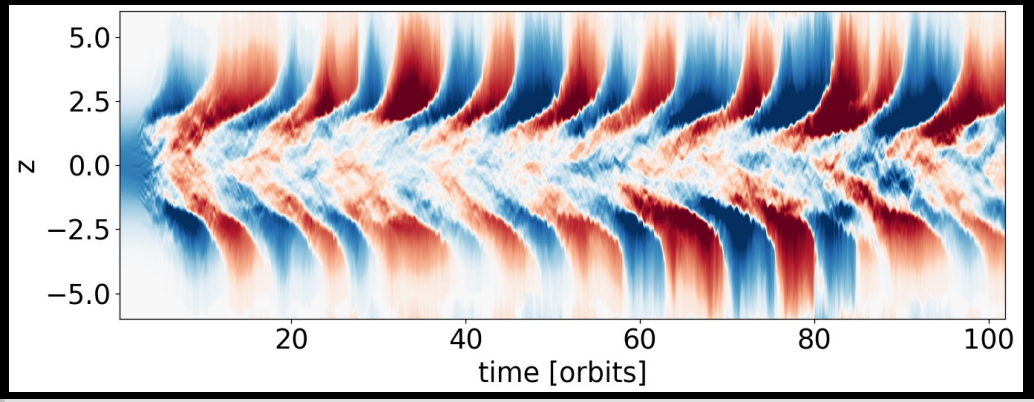
=



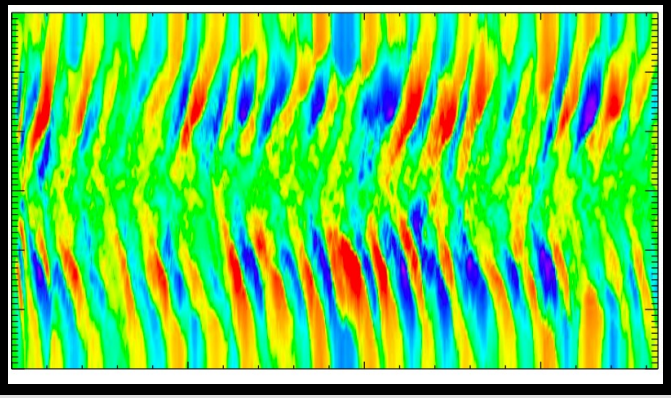
- Magnetic fields in accretion disk

First particle method to capture cyclical reversal of magnetic field for many orbits

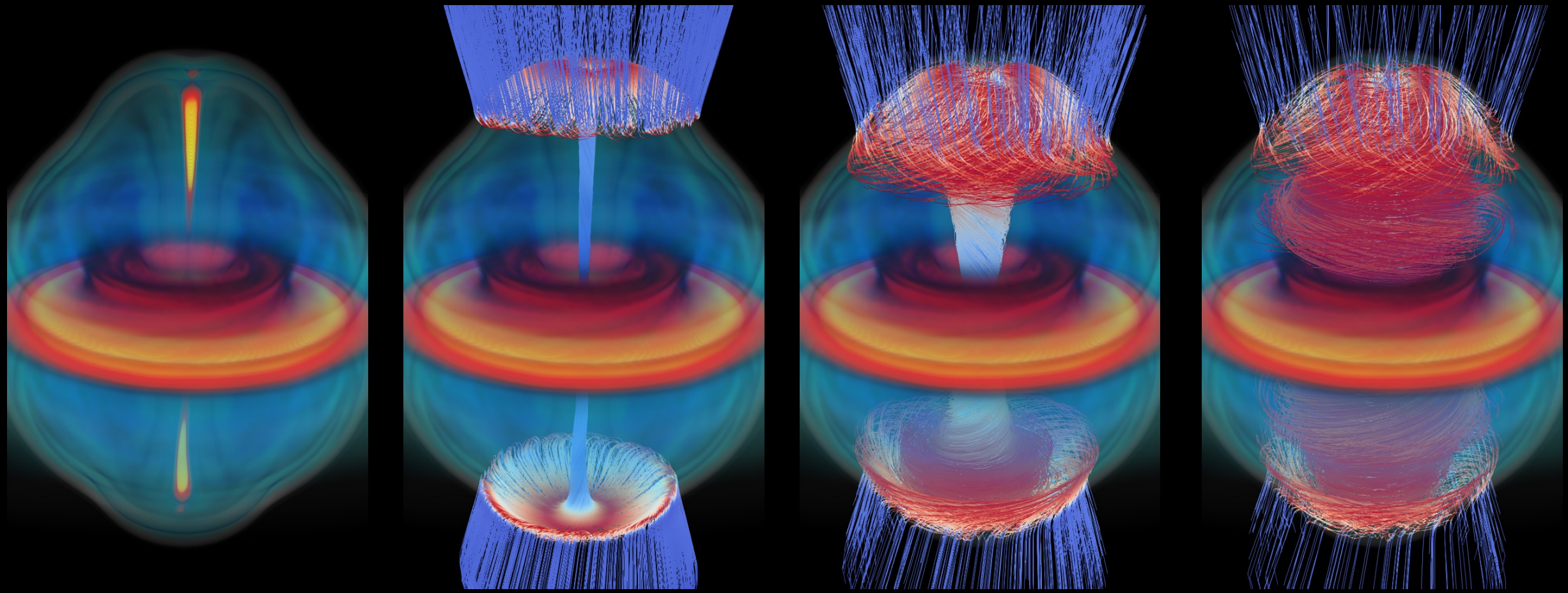




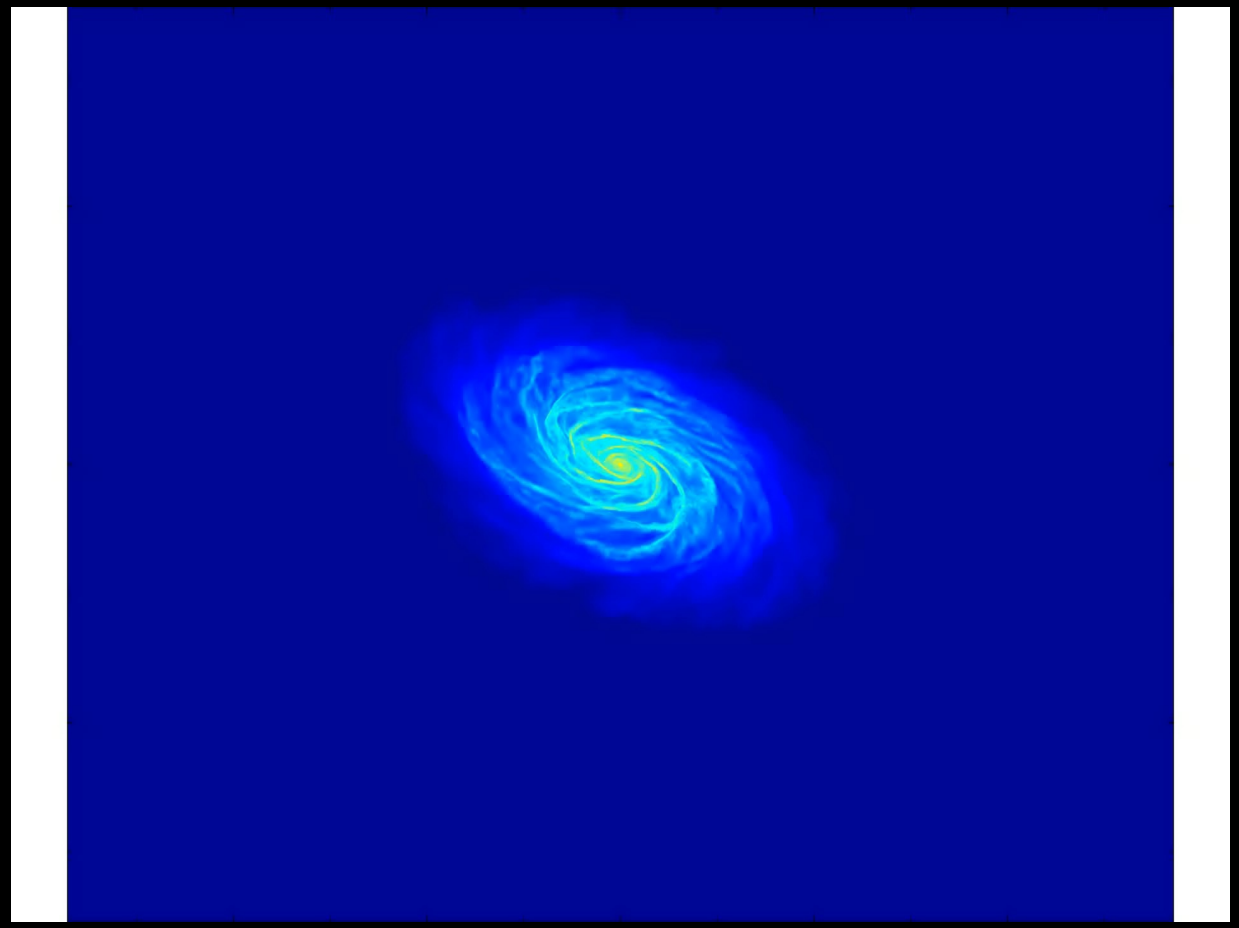
=



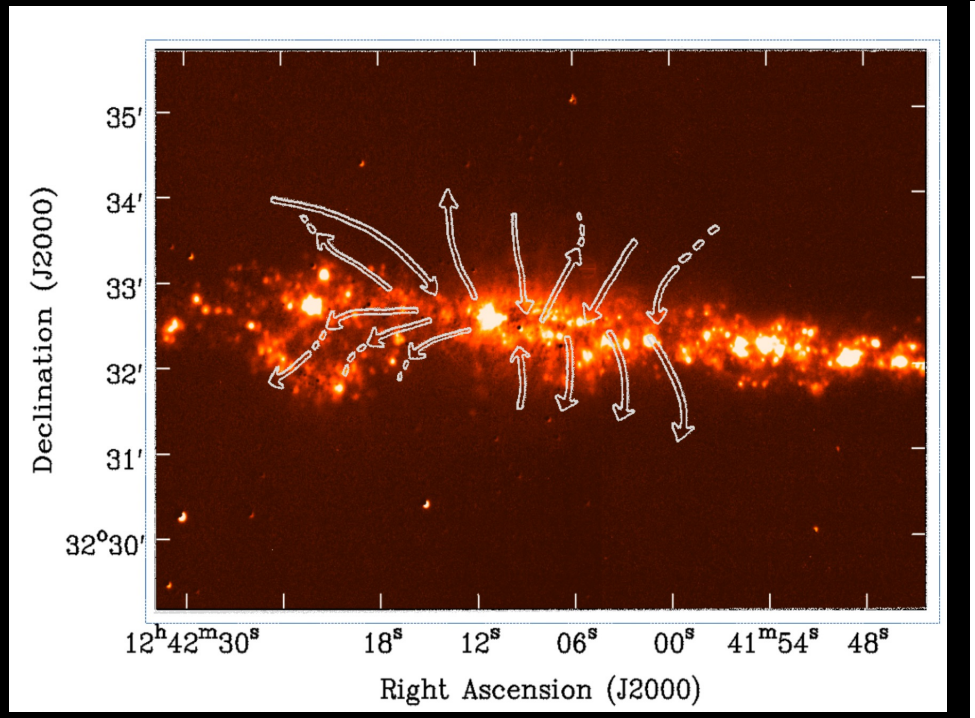
Magnetic structure around jet

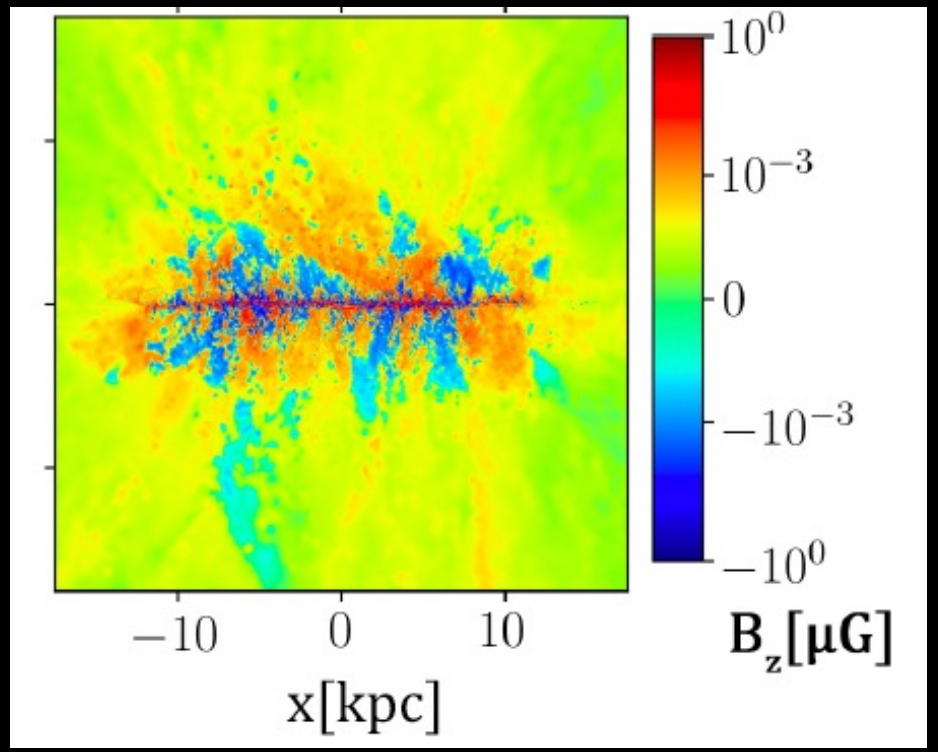


Galaxy Simulations

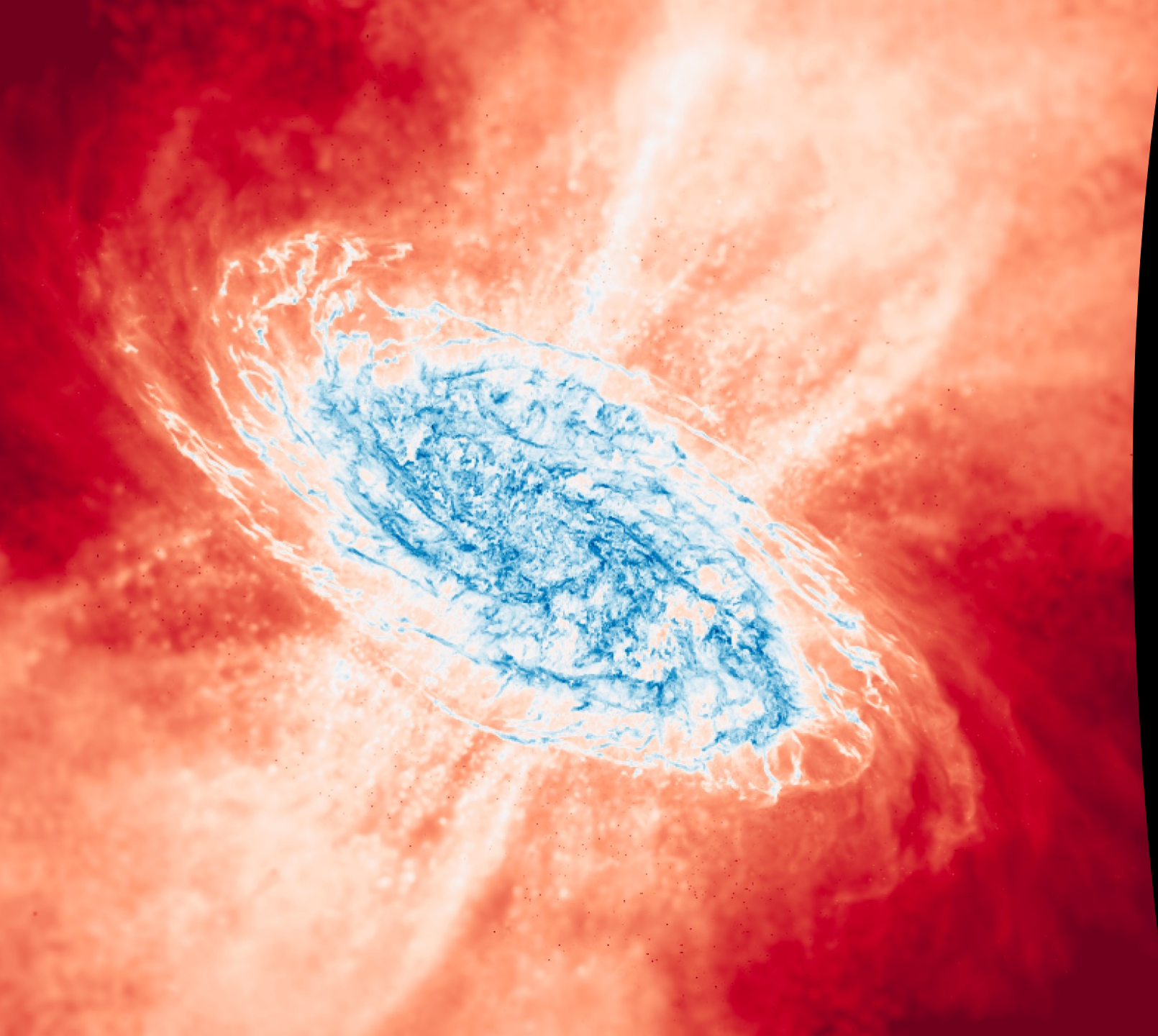


- 60 galaxy simulations with varying setup
- Magnetic field amplifies quickly, reaching saturation ~ 1.2 Gyr
- **Dynamo processes captured!**
- Strong mean-field dynamo, consistent with the classic theory of an $\alpha\omega$ dynamo.
- Magnetization of the circumgalactic medium


Declination (J2000)
35′
34′
33′
32′
31′
32°30′
12h42m30s
18s
12s
06s
00s
41m54s
48s
Right Ascension (J2000)


10^0
10^−3
0
−10^−3
−10^0
−10
0
10
x[kpc]
B_z[μG]

Beck 2019



Thank you!